

THERMAL REVISION DECK

PART 1: CORE THERMODYNAMICS & COOLING EQUATIONS

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CARD #01

THERMODYNAMICS



NEWTON'S LAW OF COOLING

Question: What fundamental equation governs the rate of heat loss of a hot core capsule submerged in a cooler ambient fluid?

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CARD #01

KEY FORMULA



EXPONENTIAL DECAY

$$T(t) = T_{\text{env}} + (T_0 - T_{\text{env}})e^{-kt}$$

Explanation: The cooling rate $\frac{dT}{dt}$ is directly proportional to the instantaneous temperature gap $(T - T_{\text{env}})$.

k = cooling constant

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CARD #02

THERMODYNAMICS



DECAY CONSTANT (k)

Question: What does the constant k quantify physically, and which capsule is better insulated: $k = 0.15 \text{ min}^{-1}$ or $k = 0.015 \text{ min}^{-1}$?

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CARD #02

INSULATION FACTOR



LOWER-k CAPSULE ($k = 0.015 \text{ min}^{-1}$)

Answer: For a lumped system, $k = \frac{hA}{mc}$ is an effective cooling constant. Under comparable conditions, a smaller k means a slower approach to the ambient temperature.

Units: min^{-1} or s^{-1}

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CARD #03

HEAT TRANSFER



FOURIER'S LAW OF CONDUCTION

Question: How does thermal energy transfer directly through solid matter, and what formula defines conduction heat flow?

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CARD #03

CONDUCTION EQUATION



MOLECULAR VIBRATION

$$\dot{Q} = -\lambda A \frac{dT}{dx}$$

Explanation: The conductive heat-transfer rate depends on thermal conductivity λ , area A and the temperature gradient. Low-conductivity materials reduce the rate.

 λ = thermal conductivity

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CARD #04

EQUILIBRIUM



THERMAL EQUILIBRIUM

Question: When does net heat transfer between the capsule core and the 0°C ice bath drop to zero? What physical law governs this?

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CARD #04

ZEROth LAW



EQUAL TEMPERATURES

$$T_{\text{capsule}} = T_{\text{env}}$$

Answer: When $T_{\text{capsule}} = T_{\text{env}}$. For the nominal ice bath this limit is 0°C , but the measured bath temperature should be used. With no temperature difference, there is no net heat transfer between them.

Net heat-transfer rate = 0

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PART 2: RADIATIVE SHIELDING & MULTI-LAYER INSULATION (MLI)

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CARD #05

RADIATION



SPACE VACUUM RADIATION

Question: Which heat-transfer mechanism can cross a vacuum between separated bodies?

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CARD #05

STEFAN-BOLTZMANN



PHOTON HEAT FLOW

$$P_{\text{rad}} = \epsilon \sigma A (T^4 - T_{\text{env}}^4)$$

Answer: Thermal radiation can cross a vacuum as electromagnetic energy. Conduction can still occur through physical contacts within a spacecraft, but not across an empty gap.

$$\sigma = 5.67 \times 10^{-8} \text{ W/m}^2 \text{K}^4$$

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CARD #06

SHIELDING



MYLAR RADIATIVE SHIELD

Question: How does a thin shiny aluminized Mylar foil protect spacecraft and astronauts from extreme solar thermal radiation?

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CARD #06

OPTICAL REFLECTION



INFRARED REFLECTANCE

Answer: An aluminized reflective film can have high infrared reflectance and low emissivity. Its performance depends on the specific product, surface condition and whether it faces a suitable gap; it is not a substitute for conductive insulation in direct water contact.

Verify product optical properties

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CARD #07

CONVECTION



CONVECTIVE AIR TRAPPING

Question: How do the sealed air cells in bubble wrap reduce heat transfer?

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CARD #07

STAGNANT CELLS



STAGNANT AIR POCKETS

Answer: Air has low thermal conductivity ($\lambda_{\text{air}} \approx 0.026 \text{ W/(m} \cdot \text{K)}$), while small sealed cells suppress bulk air circulation.

Low gas conduction + suppressed circulation

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CARD #08

MLI SPACESUITS



MULTI-LAYER INSULATION

Question: Why do spacesuits and satellites stack Cotton + Bubble Wrap + Mylar in composite layers?

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CARD #08

TRIPLE SHIELD



TRIPLE HEAT SHIELD

Answer: Composite synergy:

- **Cotton:** Blocks solid conduction.
- **Bubble Wrap:** Traps fluid convection.
- **Mylar:** Reflects infrared radiation.

All 3 heat paths blocked

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PART 3: REFERENCE MODEL, MEASUREMENTS & EXPERIMENTAL BOUNDARY

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CARD #09

SIMULATION



ANALYTICAL REFERENCE MODEL

Question: What assumptions does the SPHERE Newton cooling model make?

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CARD #09

MODEL ASSUMPTIONS



LUMPED-PARAMETER MODEL

Answer: It assumes a uniform capsule temperature, a constant bath temperature and a constant effective cooling coefficient. The model is a reference calculation, not a full digital twin.

Hardware + Simulation

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CARD #10

LAB ANALOG



ICE-WATER BOUNDARY

Question: Why is an ice-water slurry used in the comparative experiment?

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CARD #10

CONSTANT SINK



CONSTANT HEAT SINK

Answer: The slurry provides a relatively stable, repeatable cold boundary. Its actual temperature must be measured; it does not reproduce vacuum heat transfer.

 $T_{\text{env}} = 0.0^{\circ}\text{C}$ constant

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CARD #11

THERMAL MASS



HEAT CAPACITY OF WATER

Question: Why is liquid water ($c = 4184 \text{ J/kg} \cdot ^{\circ}\text{C}$) ideal for capsule core telemetry trials? What is the heat formula?

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CARD #11

THERMAL ENERGY



HIGH THERMAL MASS

$$Q = m \cdot c \cdot \Delta T$$

Answer: Water stores large thermal energy per gram, producing smooth, measurable exponential decay over 15+ minutes without dropping instantly.

 $c = 4184 \text{ J/kg} \cdot \text{K}$

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CARD #12

GRAPH ANALYSIS



DECAY CURVE CURVATURE

Question: Why is the temperature-time graph $T(t)$ steep at $t = 0$ and increasingly flat toward $t = 15 \text{ min}$?

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CARD #12

ASYMPTOTIC FLOW



SHRINKING TEMP GAP

Answer: Heat flux rate $\frac{dQ}{dt} \propto (T - T_{\text{env}})$. As the capsule cools, $(T - T_{\text{env}})$ shrinks, causing the rate of temperature drop to slow down asymptotically.

Slope = $-k(T - T_{\text{env}})$

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PART 4: EXPERIMENTAL METHOD, ERROR METRICS & SYNTHESIS

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CARD #13

METRICS

MEAN ABSOLUTE ERROR

Question: How is the mean absolute error between measured and predicted temperatures calculated?

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CARD #13

MODEL ACCURACY

MODEL RESIDUALS

$$MAE = \frac{1}{n} \sum_{i=1}^n |T_{\text{meas},i} - T_{\text{model},i}|$$

Answer: MAE is the mean magnitude of the residuals and is reported in °C. A lower value indicates closer agreement but does not, by itself, validate every model assumption.

Units: °C

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CARD #14

LAB PROTOCOL

PROBE PLACEMENT

Question: Why should the thermometer probe be positioned near the centre of the liquid rather than touching the capsule wall?

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CARD #14

BOUNDARY LAYER

BULK FLUID TEMPERATURE

Answer: Wall contact biases the reading towards the local boundary temperature rather than the bulk liquid temperature and introduces a systematic measurement error.

Use consistent probe position

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CARD #15

SI UNITS

SI UNITS REFERENCE

Question: What are the standard SI units for temperature T , thermal energy Q , thermal conductivity λ , and cooling constant k ?

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CARD #15

STANDARD UNITS

UNITS CHEAT SHEET

- **Temperature (T):** Kelvin (K) or Celsius (°C)
- **Thermal Energy (Q):** Joules (J)
- **Decay Constant (k):** s⁻¹ or min⁻¹
- **Conductivity (λ):** W/(m · K)

Standard Metric Units

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CARD #16

DESIGN COMPARISON

MULTILAYER CONFIGURATION

Question: Which tested configuration is assigned the smallest reference cooling constant, and which mechanisms may explain that choice?

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CARD #16

REFERENCE RESULT

MULTILAYER TEST ASSEMBLY

Answer: The cotton, bubble-wrap and reflective-film assembly is assigned $k = 0.015 \text{ min}^{-1}$. This is a model assumption to be evaluated against measurements, not a guaranteed material property.

Compare with measured data